TEXORA · LECTURE SHEET

Basic Weft Knitted Structures

Four structures generate almost all weft-knitted fabric: plain (jersey), rib, interlock and purl. Each is defined by how its loops face — all to one side, alternating between two beds, locked double, or alternating course by course — and that facing decides balance, curl, stretch and use.

TE-3101 · KNITTING TECHNOLOGY · TOPIC 2 · LEVEL 3 TERM 1

1. How structures are described — notation

Three records are used together. (1) **Loop diagram:** the yarn path drawn as it lies in the fabric — most truthful, slowest to draw. (2) **Needle (dot) diagram:** a grid where each dot is a needle/wale position; a cross or mark in a cell means a loop is formed there, a blank means the yarn floats or the needle misses, and face/reverse loops are distinguished by symbols (commonly × for a face loop seen from the technical face, ○ for a reverse loop). Rows of the grid are courses, columns are wales. (3) **Cam/needle arrangement:** which needles are set to knit, tuck or miss at each feeder — the machine instruction that produces the diagram. In this sheet, ● = loop drawn to the face side at that wale, ○ = loop drawn to the reverse side.

2. Plain (single jersey)

Every loop is drawn in the same direction through the previous course, on one needle bed only. The **technical face** shows the V-shaped limbs of the needle loops; the **technical back** shows the arcs of the needle-loop heads and sinker loops. Consequences: the fabric **curls** — toward the face at the top and bottom edges, toward the back at the side edges — because the bent yarn tries to straighten asymmetrically; it is the lightest and cheapest weft structure per m², has the greatest area yield from a given yarn, stretches readily (especially widthwise), and ladders from either end of a broken wale. Uses: T-shirts, single-jersey underwear, hosiery, dresses. Derivatives by needle selection: **lacoste, piqué, fleece (with inlay), terry, plated and striped jersey**.

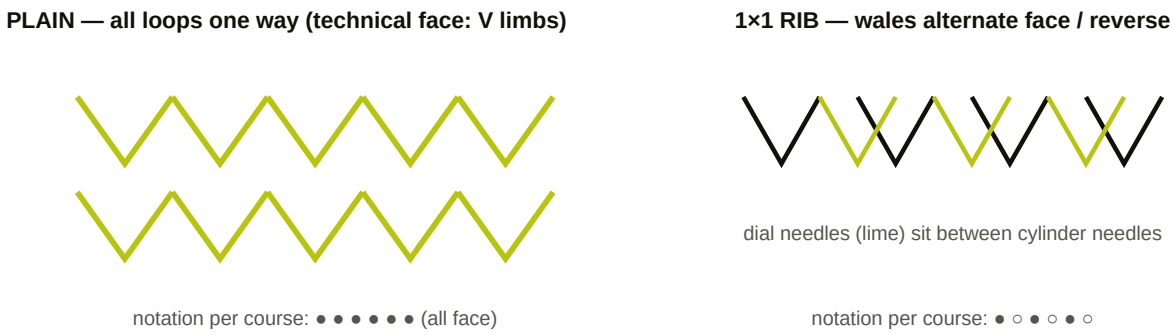


Figure 1 — Plain jersey and 1×1 rib in loop/notation form. Jersey draws every loop the same way (unbalanced, curls). Rib alternates the draw wale by wale between two beds (balanced across the width).

3. Rib structures

Rib is knitted on two beds whose needles alternate: face wales and reverse wales alternate across the width — **1×1 rib** one-and-one, **2×2 rib** two face wales then two reverse wales, and so on (written face × reverse). Because face and reverse tendencies oppose each other in the same course, rib is **balanced and does not curl**; it contracts widthwise at rest (the reverse wales hide behind the face wales) and therefore extends greatly across the width and recovers — the classic cuff, collar, waistband and neck structure. Rib is thicker and narrower than jersey from the same needles, uses more yarn per unit area, and can be unravelled only from the last-knitted end. **Half-cardigan and full-cardigan** add tuck courses to one or both beds for fuller, wider rib fabrics.

4. Interlock

Interlock is two 1×1 ribs intermeshed so that the wales of one rib sit directly behind the wales of the other. It is knitted on a cylinder-and-dial machine with two sets of needles in each bed (long and short, alternating) and two cam systems: at feeder 1 the long cylinder needles knit with the short dial needles, at feeder 2 the short cylinder with the long dial — two rib courses that together make one interlock course. Both sides of the fabric look like the face of jersey (smooth V limbs both sides), the fabric is **balanced, thick, dimensionally stable, does not curl and does not ladder readily**, extends similarly in both directions, and is warmer and more expensive than jersey. Uses: quality polos, underwear, children's wear, sportswear. Because one interlock course consumes two feeder courses, output in m/h is roughly half that of jersey at the same feeder count and speed — quantified in Topic 12.

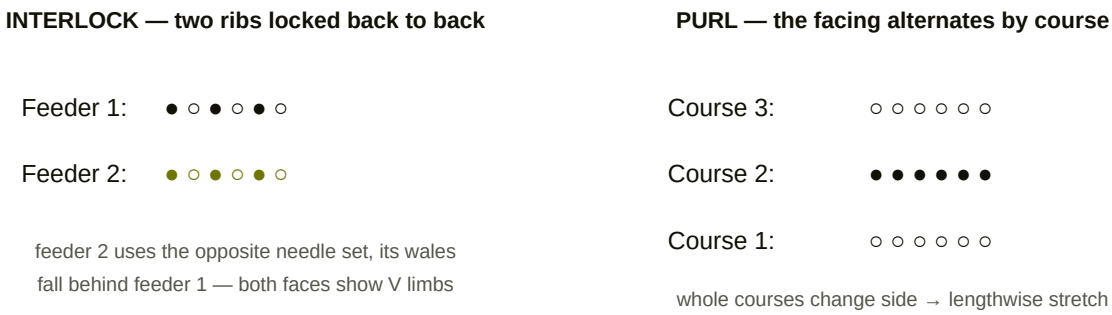


Figure 2 — Notation for interlock and purl. Interlock needs two feeders for one course; purl (links-links) changes the draw direction course by course using double-headed needles.

5. Purl (links-links)

In purl, whole courses alternate: one course is drawn to the face, the next to the reverse, using double-headed latch needles transferred between two beds in line. The fabric is balanced in the course direction, extends strongly lengthwise, and shows horizontal ridges where the facing changes. True purl machines are slow and mechanically complex, so purl survives mainly in knitwear (links-links sweater panels, garter and moss effects simulated on flat machines by transfer) rather than in bulk apparel fabric.

6. Tuck, miss and float — the three loop variants

ELEMENT	NEEDLE ACTION	EFFECT ON FABRIC
Knit loop	Needle clears fully, takes yarn, knocks over	Normal loop; the reference structure
Tuck loop	Needle rises only to tuck height (latch not cleared): new yarn is held in the hook with the old loop still on the stem	Wider, thicker, less lengthwise extension; held loop lengthens later (cardigan, piqué, honeycomb). Too many tucks on one needle overload the hook
Miss / float	Needle does not rise; yarn passes behind it as a float	Less widthwise stretch, straighter yarn path; long floats snag — jacquard backs, inlay and laid-in structures use controlled floats

7. The four basics compared

PROPERTY	PLAIN	1×1 RIB	INTERLOCK	PURL
Beds used	1	2	2 (double needle sets)	2 in line, transfer

Balance / curl	Unbalanced, curls	Balanced, no curl	Balanced, no curl	Balanced, no curl
Greatest stretch	Widthwise	Widthwise (very high + recovery)	Moderate, both ways	Lengthwise
Relative thickness/weight	Lowest	Medium	Highest of the three bulk structures	Medium
Unravels	Both ends	Last end only	Last end only, with difficulty	Both ends
Typical use	T-shirts	Cuffs, collars, bands	Polos, underwear	Knitwear effects

Glossary

Technical face/back — the side showing needle-loop limbs / the side showing loop heads and sinker loops. **Balanced structure** — face and reverse tendencies cancel, so no curl. **Tuck** — yarn held in the hook without clearing the old loop. **Float** — yarn passing unknitted behind a needle. **Cardigan** — rib with tuck courses on one (half) or both (full) beds. **Notation** — grid record of loop facing per wale and course.

Key points

• Plain = one bed, one direction: light, cheap, curls, ladders. Rib = two beds alternating by wale: balanced, widthwise recovery. Interlock = two ribs locked: smooth both sides, stable, half the length output per feeder. Purl = alternation by course: lengthwise stretch.
• Tuck widens and thickens; miss/float restrains stretch. Almost every fancy weft structure is these basics plus tuck/miss selection.

Practice

1. Draw the dot notation for 2×2 rib for one course over 8 wales, and for a purl fabric over 3 courses.
2. A polo body must look smooth on both faces, not curl at the hem, and hold shape in wear. Choose between jersey, rib and interlock and defend the choice on three properties.
3. Why does interlock need long and short needles in each bed? What fault appears if both feeders knit on the same needle set?
4. A jersey hem curls to the face at the bottom but to the back at the sides. Explain both from loop geometry, and name two practical cures used in garment making.

5. Full-cardigan fabric is wider and fuller than 1×1 rib from the same machine. Explain using the tuck element.

Answer key

1. 2×2 rib: ● ● ○ ○ ● ● ○ ○. Purl: course 1 all ○, course 2 all ●, course 3 all ○ (or the inverse) — whole rows change symbol.
2. Interlock: both faces show V limbs (smooth both sides), balanced so no hem curl, and the locked double structure resists the bagging and growth of jersey; rib would show alternating wales and jersey would curl and ladder.
3. The two feeder systems must knit opposite ribs: long needles work at one feeder, short at the next, in both beds, so the second rib's wales fall behind the first. If both feeders used the same set, the second course would knit into the same wales as an ordinary rib course — giving rib, not interlock, with the unused needles carrying no loops.
4. The yarn in each loop tries to straighten. On the free top/bottom edges the head arcs dominate and roll the fabric toward the face; at the side edges the limbs dominate and roll it toward the back. Cures: hem with a rib/interlock band or folded hem, use a balanced structure for the part, apply resin/heat-setting or a small elastane plating — any two.
5. In full cardigan every course contains tucks on both beds in turn; the held loops are not knocked over each course, so yarn accumulates and is released lengthwise into the next knit course, spreading the wales apart (wider) and adding bulk (fuller) while reducing lengthwise extension.

Texora — course notes prepared from the official syllabus of B.Sc. in Textile Engineering · Made by Sakib